

**PES University, Bengaluru**

(Established under Karnataka Act 16 of 2013)

END SEMESTER ASSESSMENT (ESA) - Jan-Feb**UE24CY151A - Engineering Chemistry****Total Marks : 100.0**

1.a. (i) What is the selection rule for an anharmonic oscillator model? What is first overtone transition in anharmonic oscillator, give its expression and explain the terms involved. **(4marks)**

(ii) If the fundamental stretching frequency of CO is at 2143 cm^{-1} and its anharmonicity constant is 0.01, calculate its oscillational frequency. **(2 marks)** (6.0 Marks)

1.b. For HBr molecule, having a bond length of 1.404 Å (angstrom), at what value of energy (in m^{-1}) will the first line appear in the rotational spectrum of the molecule. Given the Gram molar mass of H = 1.008 amu and that of Br is 79.9 amu, ($1\text{ amu} = 1.66 \times 10^{-27}\text{ kg}$, $c = 3 \times 10^8\text{ ms}^{-1}$, $h = 6.6 \times 10^{-34}\text{ Js}$) (6.0 Marks)

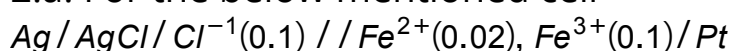
1.c. (i) Why is the phase rule applied to one component water system : $F=C-P+2$ whereas for Pb- Ag system it is $F=C-P+1$. Justify. **(3 marks)**

(ii) For the thermal decomposition of solid CaCO_3 in a closed container : $\text{CaCO}_3(s) \leftrightarrow \text{CaO}(s) + \text{CO}_2(g)$; Give the number of components and the number of phases. Justify. (6.0 Marks)

1.d. (i) What is computational chemistry? Give any three applications of computational chemistry. **(4 marks)**

(ii) During the conductometric titration of a mixture of strong and weak acid (HCl and CH₃COOH) against a strong base (NaOH), account for the nature of the conductance vs Volume of NaOH graph obtained. **(3 marks)**
(7.0 Marks)

2.a. For the below mentioned cell-



Write the anodic and cathodic reaction, E^0_{cell} and E_{cell} values. Given : The standard reduction potential values of $E^0_{Fe^{3+}/Fe^{2+}} = 0.77V$ and

$E^0_{Ag/AgCl/Cl^{-1}} = 0.28V$, $R=8.314 \text{ J/K/mol}$ and $F=96500 \text{ C/mol}$ **(6.0 Marks)**

2.b. (i) When a metal M is in contact with solution containing its ions M^{n+} and if ionization is faster than deposition, explain the formation of the electrical double layer at the interface of metal and solution. **(3 Marks)**

(ii) Derive an expression for the 'Boundary Potential' established when a glass electrode is immersed in an external solution of H^+ ions. Assume that the concentration of H^+ ions in the external solution is c_1 and the concentration of H^+ ions in the internal reference solution is c_2 . **(3 Marks)**
(6.0 Marks)

2.c. (i) If overvoltage of hydrogen gas is high on a metal surface, how does this affect the rate of corrosion of metal. (Assume that the cathodic reaction is hydrogen evolution) **(2 marks)**

(ii) Ship sailing in the sea or docked in the yard for a long time suffers from which type of corrosion. Give the anodic and cathodic regions formed and give the corresponding cathodic reactions. **(4 marks)**

(6.0 Marks)

2.d. (i) Give examples of - **(4 marks)**

- A corrosion product that offers protection
- An anodic corrosion inhibitor
- A cathodic protection technique for preventing corrosion
- The flux used in galvanization

(ii) In the potentiometric titration experiment for estimation of ferrous iron, name the indicator and reference electrode. Justify. **(3marks)**

(7.0 Marks)

3.a. (i) A battery using 15 g Zinc as anode and discharges a constant current for 2 hours. The voltage of the battery is 1.9 V and the total weight of battery is 50g. Calculate the energy density of the battery in Wh/Kg. (Given : molar mass of Zn = 65 g, $F = 96500 \text{ C/mol}$ and no. of electrons involved in the redox reaction is 2.) **(4 marks)**

(ii) Give the anodic and cathodic reactions taking place in a zinc-air battery. **(2 marks)**

(6.0 Marks)

3.b. (i) Give the Nernst equation corresponding to the output voltage generated in an 'Oxygen Sensor' and explain the terms. What are the inferences one can draw from the output voltage values and how is it useful. **(4 marks)**

(ii) Why is Polymer electrolyte membrane fuel cell operated in the temperature range of 60 -90 °C. **(2 marks)** (6.0 Marks)

3.c. (i) Give the synthesis of hydrogen using the steam reforming process with methane as the starting material. (Mention the conditions of temperature, pressure and catalyst used in the reaction). **(4 marks)**

(ii) How are 'Liquid organic hydrogen carriers (LOHC)' used in storage and transport of hydrogen gas. Give one example of LOHC. **(2 marks)** (6.0 Marks)

3.d. (i) Of all the energy storage / conversion devices discussed, give a suitable one for the following applications- **(4 marks)**

- Hearing aids
- Mobiles and laptops
- Space exploration
- In hybrid cars for start-stop application

(ii) Calculate the viscosity average molecular weight of polystyrene (atactic) from the following data: Intrinsic Viscosity =0.73, $\kappa = 11.5 \times 10^{-5}$ and Mark Houwink parameter (a) is 0.73. **(3 marks)** (7.0 Marks)

4.a. Account for the following-

- (i) Polyvinylchloride is more crystalline than polyethene. **(2marks)**
 - (ii) During the synthesis of butyl rubber, 5-10% isoprene is used. **(2marks)**
 - (iii) Bakelite and polypropylene differ completely in the way they respond to heat. **(2marks)**
- (6.0 Marks)

4.b. Give a polymer as an example for the below mentioned applications -

- (i) A polymer used to impart shrinkage/crease resistance to fabrics
 - (ii) A polymer used in making seals for vacuum applications
 - (iii) A polymer used in manufacture of aquariums and aircraft canopies.
 - (iv) A polymer used in the conductive layer of an OLED (Organic Light Emitting Diode)
 - (v) A polymer used in applications in prostheses, implants, tendon/ligament repair.
 - (v) A polymer that is a component of personal armor such as combat helmets.
- (6.0 Marks)

4.c. Account for the following

- (i) Bending of bulk Cu happens readily but nano sized Cu particles are super hard. **(2marks)**
 - (ii) Gold spheres of > 50 nm absorb at $\lambda = 575$ nm whereas gold spheres of 2-5 nm absorb at $\lambda = 517$ nm. **(2marks)**
 - (iii) Bulk gold is catalytically inactive but gold nanoparticles are catalytically active. **(2marks)**
- (6.0 Marks)

4.d. (i) Why should the use of blocking group and unnecessary derivatization be avoided during a chemical reaction. **(2 marks)**
(ii) Elaborate on 'Safer Synthesis' principle of 'Green Chemistry'. **(2 marks)**

(iii) 4 g of Ferrous ammonium sulphate (FAS) is accurately weighed out, half a test tube conc. H_2SO_4 is added to it and it is made upto mark in a 250 ml standard flask and taken in the burette. 25 ml of waste paper is mixed with acidified dichromate and this consumes 12.5 ml of the standard FAS for complete reaction in the estimation of COD of waste water sample. If the blank titre value when no waste water sample is taken is 45ml of the same standard FAS, Calculate the COD of one litre of water sample . Given the Gram equivalent weight of FAS is 392.13. **(3 marks)** (7.0 Marks)



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END SEMESTER ASSESSMENT (ESA) - Jan-Feb

UE24CY151A - Engineering Chemistry

Total Marks : 100.0

- 1.a. i) Why HI molecule is IR active but H_2 and I_2 are IR inactive? (1M)
- ii) The first spectral line in the rotational spectrum of HI molecule appears at 13.022 cm^{-1} . Calculate the bond length. (3M)
(Given: $c = 3 \times 10^8\text{ m/s}$, $1\text{amu} = 1.66 \times 10^{-27}\text{ kg}$, gram molar mass of H = 1.0 amu and I = 126.9 amu, $\pi = 3.14$, Planck's constant = $6.627 \times 10^{-34}\text{ Js}$)
- iii) Draw an energy level diagram of ground and excited electronic energy states of a diatomic molecule and show vibrational and rotational energy levels within them. (2M) (6.0 Marks)

- 1.b. i) Using harmonic oscillator model calculate zero point vibrational energy of NO molecule given that the force constant of NO is 1530 N/m . (4M)
- ii) Draw the vibrational energy level diagram for the above model and draw the corresponding spectra obtained from the transition. (2M)
- iii) Calculate the energy of $v=3$ and $v=4$ level. (1M)
(Given: $c = 3 \times 10^8\text{ m/s}$, $1\text{amu} = 1.66 \times 10^{-27}\text{ kg}$, gram molar mass of N = 14.0 amu and O = 16 amu, $\pi = 3.14$, Planck's constant = $6.627 \times 10^{-34}\text{ Js}$) (7.0 Marks)

- 1.c. i) Draw a neat labeled phase diagram of water system (mention the areas, lines, triple point, critical point, pressure, temperature) and calculate the degree of freedom along the freezing point curve. (4M)
- ii) What is condensed phase rule and why is it applied to the two component lead-silver system? (2M) (6.0 Marks)

1.d. Derive phase rule for a system of C components and P phases which are in equilibrium with each other. (4M) (4.0 Marks)

1.e. i) Mention any two factors affecting the conductance. (1M)
ii) Write any two advantages of potentiometric titration. (1M)
(2.0 Marks)

2.a. Derive Nernst equation for the given cell and calculate the emf of the cell.
 $2Al + 3Cu^{2+}(0.6M) \rightarrow 2Al^{3+}(0.35M) + Cu$
(Given: $E^0_{cell} = 2.0V$) (6.0 Marks)

2.b. i) What is ion selective electrode? Write the cell representation when a glass electrode is coupled with a saturated calomel electrode. (2M)

(ii) When a glass electrode was coupled with a saturated calomel electrode to measure the pH of an unknown solution, an emf of 0.786V was recorded at 298K when pH = 4.2. At the same temperature when a solution of unknown pH was used, an emf of 0.825V was recorded. Calculate the pH of the unknown solution. (Given: $E_{SCE}=0.2412V$) (3M)

(5.0 Marks)

2.c. Two metals such as iron and tin are in contact with each other and exposed to the corrosive environment, name the type of corrosion and explain with the relevant reactions. (5.0 Marks)

2.d. Account for the following:

i) Influence of hydrogen over voltage on the rate of corrosion (2M)

ii) Anodic corrosion inhibitors (2M)

iii) Sacrificial anode method (2M) (6.0 Marks)

- 2.e. i) Why colorimetric estimation of copper is done at 620 nm? (1M)
ii) Draw the nature of the graphs in the determination of pKa value of weak acid. (2M) (3.0 Marks)

3.a.

For a Zn-air battery

- i) What are the materials used as anode, cathode, electrolyte and separator? (2M)
ii) Write the reactions taking place at anode and cathode when the battery is discharging. (2M)
iii) Define energy density of a battery and calculate energy density of zinc-air battery in Wh/kg. If 6.6g of Zn is stored in the battery weighing 36.8g. (Given; voltage of the given Zinc-air battery is 1.3 V and gram molar mass of zinc is 65.38) (3M) (7.0 Marks)

3.b. i) Explain the construction and working of H₂-O₂ alkaline fuel cell. (4M)

- ii) Calculate the efficiency of the H₂-O₂ alkaline fuel cell. (Given; the enthalpy of formation of water is -285.83 kJ/mole, E_{cell}=1.19V). (2M) (6.0 Marks)

3.c. i) Mention the advantages and Limitations of Steam reforming method of hydrogen production. (2M)

ii) "Liquid organic hydrogen carriers (LOHC) can be used as storage media for hydrogen" Justify the statement with example. (2M) (4.0 Marks)

3.d. What is an oxygen sensor? Give the construction, working of oxygen sensor and explain its role in automobiles. (6.0 Marks)

3.e. Calculate the viscosity average molecular weight of a polymer solution by the following data.

$K=45.3 \times 10^{-5}$, $a=0.64$, intrinsic viscosity=0.75 (2.0 Marks)

4.a. Account for,

i) High strength of kevlar fibre. (2M)

ii) High electrical conductance of conducting polyaniline. (4M)

Explain the above with suitable structures. (6.0 Marks)

4.b. i) Give the synthesis of carbon fibre from polyacrylonitrile (with reactions). (4M)

ii) Calculate the weight average molecular weight of a polymer having 10 molecules of molecular weight is 3000, 15 molecules of molecular weight is 3500 and 25 molecules of molecular weight is 4000. (2M)

(6.0 Marks)

4.c. i) Describe the construction and working of OLEDs. (4M)

ii) Write any two principles of green chemistry. (2M) (6.0 Marks)

4.d. Give reasons for the following:

- i) Spatial confinement affects the electrical conductivity of nanomaterials. (2M)
- ii) Bulk gold is yellow in colour whereas gold nanoparticles are red in colour. (2M) (4.0 Marks)

4.e. i) 50 mL of sewage sample of COD is reacted with 30.0 mL of potassium dichromate solution and the unreacted potassium dichromate requires 12.0 mL of 0.5N FAS solution. Under similar conditions, in blank titration 20.0 mL of FAS is used up. Calculate the COD of the sample. (2M)

- ii) Name the salts causing permanent hardness. (1M) (3.0 Marks)



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END SEMESTER ASSESSMENT (ESA) - JAN 2023

UE23CY151A - Engineering Chemistry

Total Marks : 100.0

1.a. For a rigid rotor diatomic molecule:

(i) Draw the energy level diagram upto $J = 5$.

(ii) Draw the rotational spectrum .

(iii) Calculate the moment of inertia and the distance between two spectral lines in the spectra if $B = 10.56 \text{ cm}^{-1}$.

(Given : $c = 3 \times 10^8 \text{ m/s}$, $\pi = 3.14$, $h = 6.626 \times 10^{-34} \text{ Js}$ respectively)

(6.0 Marks)

1.b. (i) The force constant of HF molecule showing simple harmonic oscillation at 970 N/m . Calculate the oscillation frequency (cm^{-1}) and zero point energy.

(ii) Why does the spectrum shows only one line in simple harmonic oscillation ?

(Given: $c = 3 \times 10^8 \text{ m/sec}$, $\pi = 3.14$, $N = 6.02 \times 10^{23}$, $1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$, atomic mass of $\text{H} = 1.008 \text{ amu}$, $\text{F} = 18.99 \text{ amu}$).

(6.0 Marks)

1.c. Derive phase rule for a system of C components and P phases which are in equilibrium with each other. (4.0 Marks)

1.d. i) Draw a neat labeled phase diagram of Lead-Silver system indicating the different areas and lines. Give the temperature and composition values corresponding to eutectic point.

ii) For Fe- C system, mention the phases that are in equilibrium with each other, corresponding temperature and weight % Carbon at the Peritectic point. (6.0 Marks)

1.e. (i) Mention the factors affecting the conductance of a solution.

(ii) In conductometric titration experiment, 25 cm³ acid mixture titrated against 1M NaOH and the graph reveals V₁ and V₂ are 2.8 cm³ and 6.5 cm³ of NaOH respectively. Calculate the normality of hydrochloric acid and acetic acid. (3.0 Marks)

2.a. For the cell:

Pt/Hg/Hg₂Cl₂/Cl⁻(0.1N)//Fe³⁺(0.05), Fe²⁺(0.5) /Pt

i) Write the anodic, cathodic and overall reaction.

(ii) Calculate E^o_{Cell} and E_{Cell} at 25 °C

(Given: E^o_{calomel} = 0.281V, E^o_{Fe³⁺/Fe²⁺} = 0.77 V). (6.0 Marks)

2.b. (i) What is ion selective electrode? Write the cell representation when a glass electrode is coupled with a calomel electrode.
(ii) When a buffer of pH = 4.0 is used an emf of 0.786V is recorded at 298K. At the same temperature when a solution of unknown pH is used an emf of 0.825 V is recorded. Calculate the pH of the unknown solution. (Given: $E_{SCE} = 0.2422$ V) (6.0 Marks)

2.c. i) Explain the effects of following factors on corrosion of metal
a) Nature of corrosion product b) Hydrogen overvoltage
ii) Identify the type of corrosion taking place, when steel pipe connected with copper plumbing. (5.0 Marks)

2.d. i) Explain the process of anodizing with the help of a diagram and relevant reaction.
ii) How is sealing done for anodized articles? (6.0 Marks)

2.e. (i) Mention the role of sulphuric acid in the estimation of ferrous iron by potentiometric titration method.
ii) Why pH is equals to pKa at half equivalence point in the pH metric titration of weak acid vs strong base ? (2.0 Marks)

3.a. i) what are reserve batteries? Give an example.
ii) Calculate the capacity (Ah), energy density (Wh/kg) and power density (W/kg) of a battery if 2.4 g of Lithium is stored in the battery and the battery lasts for 120 mins. The battery weighs 115 g and gives steady voltage of 3.4 V. (Given: $F = 96500 \text{ C/mol}$, atomic mass of lithium = 7) (6.0 Marks)

3.b. i) With the help of neat labeled diagram explain the construction and working of $\text{H}_2\text{-O}_2$ alkaline fuel cell.
ii) Calculate the potential of the $\text{H}_2\text{-O}_2$ alkaline fuel cell, if its efficiency is 84.25 %. (Given the enthalpy of formation of water is -285.83 kJ/mole). (6.0 Marks)

3.c. i) Mark the regions in which the following energy storage devices appear in the Ragone plot.

a) Fuel cell (b) Li-ion battery (c) Supercapacitor

ii) Mention any two materials used for hydrogen storage. (6.0 Marks)

3.d. i) Discuss the working principle of glucose biosensor with relevant reactions.

ii) Mention any one limitation of glucose sensor. (5.0 Marks)

3.e. 50 cm³ of given hard water sample requires 15.6 cm³ of 0.075 M EDTA for complete reaction in the estimation of hardness by complexometry. Calculate the total hardness of water. (2.0 Marks)

- 4.a. i) Explain how the structure of polymer affects its crystallinity.
ii) What are the monomers used in the synthesis of Butyl rubber?
iii) Write any two principles of Green chemistry. (6.0 Marks)

- 4.b. i) Give the synthesis of kevlar from 1, 4 phenylene diamine.
ii) "Kevlar is 5 times stronger than steel", justify the statement.
iii) Give any one application and mention any one disadvantage. (6.0 Marks)

- 4.c. Calculate the number average, weight average and viscosity average molecular weight of a polymer having 10 molecules of molecular weight 3000, 15 molecules of molecular weight 3500 and 25 molecules of molecular weight 4000. (Given : $a = 0.65$). (6.0 Marks)

4.d. The optical properties of nano materials change with size of the nanoparticles. Explain using

- i) Quantum confinement ii) Surface plasmon resonance (4.0 Marks)

4.e. i) Define viscosity co-efficient.

ii) Write expression of Lambert- Beer's law and explain the terms.

(3.0 Marks)



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END SEMESTER ASSESSMENT (ESA) - JAN 2023

UE23CY151A - Engineering Chemistry

Total Marks : 100.0

1.a. For a rotating diatomic molecule,
i) Derive an expression for moment of inertia 'I' for the molecule.
ii) If $B = 10 \text{ cm}^{-1}$, give the rotational energy levels of this molecule, in cm^{-1} for rotational quantum numbers $J=0,1,2,3$, and draw an energy level diagram for the same molecule up to $J = 3$. (6.0 Marks)

1.b.

i) The radiation absorbed by CO during a vibrational transition is 2168 cm^{-1} . Calculate ground state vibrational energy and force constant. (Use simple harmonic oscillator model,

Given: Avagadro's number = 6.023×10^{23} , speed of light = $3 \times 10^{10} \text{ cm/s}$, $1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$ gram molar mass of C = 12 and O = 15.999, $\pi = 3.14$, Planck's constant = $6.627 \times 10^{-34} \text{ Js}$)

ii) Only one peak is observed in the simple harmonic oscillator model. Substantiate.

(6.0 Marks)

1.c. (i) Draw a neat labeled phase diagram of the Carbon dioxide system specifying the different areas and lines. Give temperature and pressure values at the triple point and critical point.
(ii) What is the Reduced phase rule? Why is general Gibb's Phase rule not applied to the 2-component system? (6.0 Marks)

1.d.

(i) Define the term component. Mention the number of phases and components for the following closed system



(ii) Define the following terms Eutectoid and Peritectic points in the Fe-C system.

(4.0 Marks)

1.e. (i) What are the factors affecting the conductance of the electrolyte solution?

(ii) Mention the unit for conductance measurement.

(3.0 Marks)

2.a.

A decinormal calomel electrode is used to determine the potential of the redox electrode Pt/Cu^{2+} (0.58 M), Cu^+ (0.08M). (Given: $E^\circ_{\text{Hg}/\text{Hg}_2\text{Cl}_2/\text{Cl}^-}=0.242\text{V}$, $E_{\text{Cu}^{2+}/\text{Cu}}=0.153\text{V}$).

1. Write cell representation
2. Write the redox reactions at electrodes
3. Calculate E°_{cell} and E_{cell} at 298 K.

(5.0 Marks)

2.b.

- i) What is the asymmetric electrode potential for a glass electrode?
ii) A glass electrode dipped in a solution of pH of 4 offers an EMF of 0.2066 V with SCE at 298 K. Dipped in a solution of unknown pH, at the same temperature, in contact with SCE, recorded an EMF of 0.1076 V. Calculate the pH of the solution if ESCE is 0.2412 V

(6.0 Marks)

2.c. (i) Consider two situations: one where Iron (Fe) is in contact with copper (Cu) and the other where iron is in contact with silver (Ag). Given the standard reduction potentials of Fe, Cu, and Ag as -0.44 V, 0.34 V, and 0.8 V respectively, determine which scenario would lead to a faster corrosion of Fe and explain the reason.

(ii) If the metal ions liberated at the anode accumulate in the anodic region, what impact would this have on the corrosion rate? Explain the observed effect.

(iii) Describe the conditions under which oxides of antimony and arsenic can be employed for corrosion control. Explain how these oxides contribute to a reduction in the corrosion rate.

(6.0 Marks)

2.d.

1. What is anodizing of aluminum? How can one produce anodized aluminum articles?
2. How is sealing done during anodization, Give the relevant reaction.
3. How can colored anodized articles be produced?

(6.0 Marks)

2.e. What are the two factors on which the absorbance of light of a particular wavelength by a substance in solution varies with? (2.0 Marks)

3.a. i) Give any two reasons why secondary batteries have limited cycle life.
ii) A battery using 1.7 g of lithium as anodic material gives a steady voltage of 3.6 V and lasts for 2 hours. If the weight of the battery is 50 g, then calculate the power density and energy density of the battery. [Given: Atomic mass of lithium is 7, $F = 96500 \text{ C/mol}$] (6.0 Marks)

3.b. i) If the E Cell for H -O alkaline fuel cell is 1.23 V the heat evolved during the reaction $\Delta H = -285.8 \text{ kJ/mole}$, calculate the efficiency of the fuel cell.
(ii) Why can't carbon-containing fuels be used in an alkaline fuel cell?
(iii) Porous carbon materials are used as electrode materials in supercapacitors. Give reason. (6.0 Marks)

3.c. i) What is a biosensor? Mention the components required for the construction of glucose sensors.

ii) Give any two points of difference between a potentiometric and amperometric sensor. (6.0 Marks)

3.d. Mention the five different types of materials used for hydrogen storage. (5.0 Marks)

3.e. Give any two advantages of the bottom-up solution combustion synthesis method for the synthesis of nanomaterials. (2.0 Marks)

4.a.

Discuss the structure-property relationship with regard to the following properties of polymers (i) Stereoregularity (ii) Tensile strength

(6.0 Marks)

4.b. Give the synthesis of epoxy resin by the polymerization of monomers. How does curing impart a high degree of hardness to the polymer? Give any two applications of epoxy resin. (6.0 Marks)

4.c.

Complete the following table for OLEDs :

Anode	
Cathode	
Emissive layer	
Conductive layer	
Advantage	
Application	

(6.0 Marks)

4.d.

(i) When the particle size is of nano order there is a decrease in the conductance of the species when compared to bulk materials. Give reasons.

(ii) State any one principle of Green Chemistry and give relevant examples.
(4.0 Marks)

4.e. 25 ml of sewage water sample is mixed with 25 ml of K Cr O solution and the unreacted K Cr O consumed 9 ml of 0.25N FAS to give the colour change with ferroin indicator. Under similar conditions, in blank titration 15ml of FAS is used up. Calculate the COD (Chemical Oxygen Demand) of the sample in mg/L (ppm).
(3.0 Marks)

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END SEMESTER ASSESSMENT (ESA) - Feb 2026**UE25CY151A - Engineering Chemistry****Total Marks : 100.0**

1.a. The first rotational line in $^{12}\text{CO}^{16}\text{O}$ is observed at 3.84235 cm^{-1} and that of $^{13}\text{CO}^{16}\text{O}$ is observed at 3.67337 cm^{-1} . Calculate the atomic mass of the ^{13}C assuming that the mass of ^{16}O to be 15.9949 amu and ^{12}C is 12.0000 amu. Assume that the bond length is same in both cases. (6.0 Marks)

1.b.

1. Why do we obtain a single line in the simple harmonic oscillator (SHO) model?
2. What is zero-point energy?
3. How can you obtain the expression for the fundamental absorption band, first overtone, and second overtone using the anharmonic oscillator model?
4. State the Franck-Condon principle.

(6.0 Marks)

1.c.

1. Using Gibbs' phase rule, justify why the degree of freedom becomes zero at the triple point of a one-component water system.
2. Under what thermodynamic constraints is the reduced phase rule applied to a two-component system, and why is it necessary to use the reduced form instead of the general phase rule?
3. For the Pb-Ag binary phase diagram, determine the degrees of freedom along the liquidus (melting-point depression) curves and at the eutectic point, clearly stating the assumptions involved.

(6.0 Marks)

1.d. What is computational chemistry? What are the steps involved in the computational determination of electronic structure? (4.0 Marks)

1.e. Justify the selection of 620 nm for the colorimetric estimation of copper. Discuss the role of ammonia in the colorimetric determination of copper and name the complex formed. (3.0 Marks)

2.a. For the Calomel Electrode:

- i) Draw a neat, well-labelled diagram.
- ii) Write the working principle.
- iii) With respect to which ion is the calomel electrode reversible?
- iv) State the advantages of the calomel electrode. (6.0 Marks)

2.b. i) For a glass electrode, What is **asymmetric potential and alkaline error**?

ii) A cell is formed by dipping a nickel rod in **0.01 M Ni²⁺** solution and a lead rod in **0.5 M Pb²⁺** solution. Write the **cell representation, cell reaction**, and **calculate the EMF of the cell**. {Given: F= 96500 C/mole, T= 298K, R=8.314 J mol⁻¹K⁻¹} $E_{Cell}^0 = 0.11V$ (7.0 Marks)

2.c. According to the electrochemical theory of corrosion, when does liberation of hydrogen and absorption of oxygen occur? Explain with suitable reactions. (5.0 Marks)

2.d. **Account for the following:**

i) Galvanic corrosion is better explained by the **galvanic series** than by the **electrochemical series**.

ii) The use of an insufficient quantity of **anodic corrosion inhibitors** is more disastrous than not using them at all. (4.0 Marks)

2.e. In the estimation of Fe^{2+} ions by potentiometric titration, the volume of FAS taken is 25 mL, and the volume of 0.043 N $\text{K}_2\text{Cr}_2\text{O}_7$ required for the reaction is 5.5 mL. Calculate the amount of iron present in 1000 mL of its solution. (Given: gram equivalent weight of iron = 55.85). (3.0 Marks)

3.a. Explain the construction and working of Li-ion batteries (along with discharging equations). (6.0 Marks)

3.b. Explain the construction and working of a supercapacitor (5.0 Marks)

3.c. Describe the components and working of a glucose biosensor
(6.0 Marks)

3.d. Briefly explain the following two material-based methods for solid-state hydrogen storage. Provide one example for each (1)Complex hydride and (2) Chemical hydrogen
(4.0 Marks)

3.e. (1) Calculate the emf of $\text{H}_2\text{-O}_2$ alkaline fuel cell.
[Given: Efficiency of the cell = 80.20 %, $\Delta H = -285.8 \text{ kJ/mole}$]
(2) In the determination of the pKa value of a weak acid by pH metric titration, what happens at half equivalence?
(4.0 Marks)

4.a. (1) Describe the synthesis and properties of Kevlar. (4M)
(2) Give any two advantages of solution combustion synthesis (2M)
(6.0 Marks)

4.b. Explain the structure–property relationship of polymers. Briefly discuss the effect of crystallinity, configuration, and tensile strength of polymers. (6.0 Marks)

4.c. Explain the four fundamental reasons why nanomaterials exhibit physical and chemical properties that deviate significantly from their bulk counterparts. Illustrate this deviation with one specific example of a change in optical properties (blue shift). (5.0 Marks)

4.d. (1) What are conducting polymers, and what are the oxidizing agents used for oxidative doping of the conducting polymer? (2M)
(2) How is Graphene Oxide (GO) converted into Reduced Graphene Oxide (rGO)? (2M)
(3) Define "Atom Economy" in the context of Green Chemistry. (1M)
(5.0 Marks)

4.e. How does the energy gap between the Highest Occupied Molecular Orbital (HOMO) and the Lowest Unoccupied Molecular Orbital (LUMO) dictate the selection of organic molecules for specific color emission in OLEDs?
(3.0 Marks)